

RISHIK MUTHYALA

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Education

University of Massachusetts Amherst

Sep. 2023 – May 2026

Bachelor of Science in Computer Science & Mathematics, Minor in Business — GPA: 3.9/4.0

Amherst, Massachusetts

Relevant Coursework

Core: Data Structures, Algorithms, AI/Machine Learning, Object Oriented, Discrete Mathematics, Probability & Statistics, Calc I/II, Linear Algebra, Multivariate Calculus, Differential Equations, Accounting, C Programming

Experience

SelfServe

January 2025 – Present

iOS Developer

Boston, MA

- Architecting native iOS application using SwiftUI and Combine framework with MVVM architecture; developed custom UI components and implemented CoreData persistence layer, reducing development time-to-market by 40%
- Designed serverless backend infrastructure using AWS Lambda, API Gateway, and PostgreSQL, implementing WebSocket connections for real-time analytics and guest management capabilities across 100+ hospitality properties

Treevah

December 2024 – Present

Software Development Engineer Intern

Chicago, Illinois

- Engineering a scalable file management system using React.js and TypeScript, implementing B-tree based hierarchical subfolder organization with Redis caching, resulting in $O(\log n)$ retrieval complexity and 40% improvement in file access latency
- Developing an intelligent search functionality using Elasticsearch and Redux for state management, incorporating fuzzy matching algorithms and contextual menus with WebSocket-powered real-time updates, reducing average file access time by 25%
- Architecting cloud infrastructure using Azure App Service, Azure Functions, and Azure Key Vault, implementing JWT-based authentication, RESTful API endpoints with rate limiting, and implemented event-driven services using Azure Service Bus for scalable data operations
- Leading AI integration research leveraging OpenAI's GPT-4 and Azure Cognitive Services; developed proof-of-concept using transformer models for automated file categorization and metadata extraction, demonstrating 60% reduction in manual organization tasks

Drobot Co.

May 2023 – September 2023

Software Engineer Intern

Boston, MA

- Developed autonomous drone prototype using Python, TensorFlow, and PyTorch, improving object tracking accuracy by 30% and implemented real-time MAVLink communication system for seamless data transfer between control station and drone
- Optimized mapping algorithms, reducing processing time by 40% and designed custom object detection model accuracy for multi-object tracking
- Utilized CAD for drone component prototyping, reducing weight by 25% and increasing flight time by 15%
- Created data analytics pipeline with Pandas and NumPy, processing 1TB+ of flight data; integrated sensor fusion algorithms, enhancing drone stability and precision by 35%

Projects

FoundU (2024 Hackathon Winner) | React Native, Node.js, Express, Google Vision API, Puppeteer November 2024

- Engineered lost UCard recovery system serving 30,000+ UMass students, enabling finders to photograph cards, with Google Vision API extracting student information and custom Puppeteer automation scraping UMass finder directory
- Developed streamlined user flow using React Native for cross-platform interface, Node.js/Express backend for secure data processing, and Nodemailer for automated location-based email notifications

Leadership

First Tech Robotics

June 2022 – June 2023

Co-Captain

Woonsocket, RI

- Co-captained and programmed Java-based competitive robot, implementing sensor fusion and PID algorithms to optimize autonomous task execution in dynamic environments

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, HTML, CSS **Tools:** VS Code, Xcode, Android

Studio, Git, GitHub, Docker, Arduino, Linux, Figma

Frameworks/Libraries: React, React Native, Node, Express, Next, SwiftUI, TensorFlow, PyTorch, Redux, Pandas, NumPy, WebSockets, OAuth, Azure SDK, Puppeteer

Cloud/Database: AWS, GCP, Azure, MongoDB, SQL, PostgreSQL